

#### 4.SVM-kernel Classification

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.76      | 0.96   | 0.85     | 85      |
| 1            | 0.88      | 0.47   | 0.61     | 49      |
| accuracy     |           |        | 0.78     | 134     |
| macro avg    | 0.82      | 0.72   | 0.73     | 134     |
| weighted avg | 0.81      | 0.78   | 0.76     | 134     |

1. How will you evaluate the model?

- Precision
- Recall
- F1-score
- Support
- accuracy

2. What does Precision mean?

Out of all records predicted as Purchased, how many are actually Purchased?

In this purchase prediction model predicts 100 people purchased only 88 people got purchased.

3. What does recall means?

Out of all actual purchased, how many did the model identify?

Only 47% of actual Purchased samples were detected.

4. What is F1 Score?

It balances Precision and Recall.

Although Precision is high, Recall is poor, so F1-score is also relatively low.

5. Why is Accuracy?

Accuracy is the percentage of total predictions that the model classified correctly.

6. Can we consider Accuracy alone for find the good mode?

No, accuracy alone cannot determine whether a model is good. It works well for balanced datasets, but for imbalanced datasets it can be misleading. For example, in fraud detection, a model may achieve 99% accuracy by predicting every transaction as non-fraud, yet fail to detect any fraudulent transactions. Therefore, we should also evaluate Precision, Recall, F1-score, ROC-AUC, and the Confusion Matrix based on the business objective.